

Phase 2 Exit Gate Review

Requirements Phase (Design Inputs) · CSI-100 Cell Sorting Instrument (IVD)

Item	Detail
Decision requested	Exit Requirements phase; start Development
Review date	22 January 2026
Document	PGR-CSI-G2 Rev A
Presenter	Atif Lodhi, Systems Engineering
Chair	VP Research and Development

Are the design inputs ready for detailed design?

Phase 0 Concept Exited Dec 2024	Phase 1 Feasibility Exited May 2025	Phase 2 Requirements Exit gate today	Phase 3 Development Exit Aug 2026	Phase 4 Verification Exit Feb 2027	Phase 5 Validation Exit May 2027	Phase 6 Transfer Exit Jun 2027
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Go means

The requirements baseline is frozen under change control and detailed design starts.

No-go means

A criterion is not met or an open item blocks design; the phase continues.

How to read this deck

Each evidence slide names the exit criteria it covers and their status, top right.

9 met · 5 met with action · 0 not met

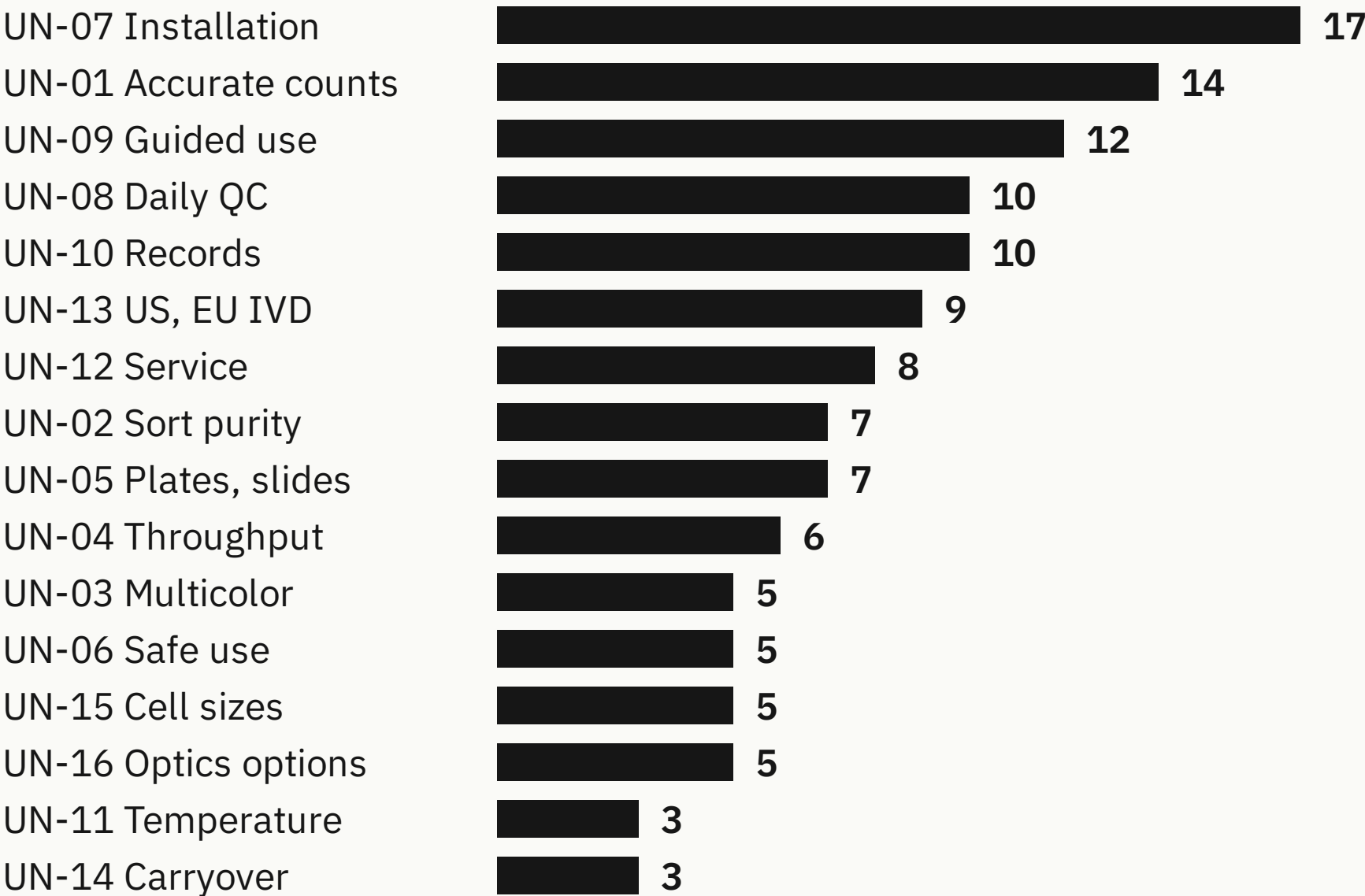
1 User needs baselined Met Slide 4	2 Requirements approved, stable Met Slides 4, 10	3 Source, allocation, method on each Met Slide 6	4 Quality review closed Action A2 Slide 5	5 Needs traced to requirements Met Slide 6	6 Software flow- down Action A1 Slide 6	7 Architecture, interfaces Action A5 Slide 7
8 Software safety class Action A3 Slide 7	9 Hazards and risk controls Action A4 Slide 8	10 Regulatory pathway Met RS-CSI-001	11 Verification strategy Met VVS-CSI-001	12 Program risks owned Met Slide 9	13 Gate 1 actions closed Met 5 of 5	14 Resources, schedule Met PMP-CSI-001

Met Met with action: satisfied for detailed design; remaining work owned and due before Gate 3

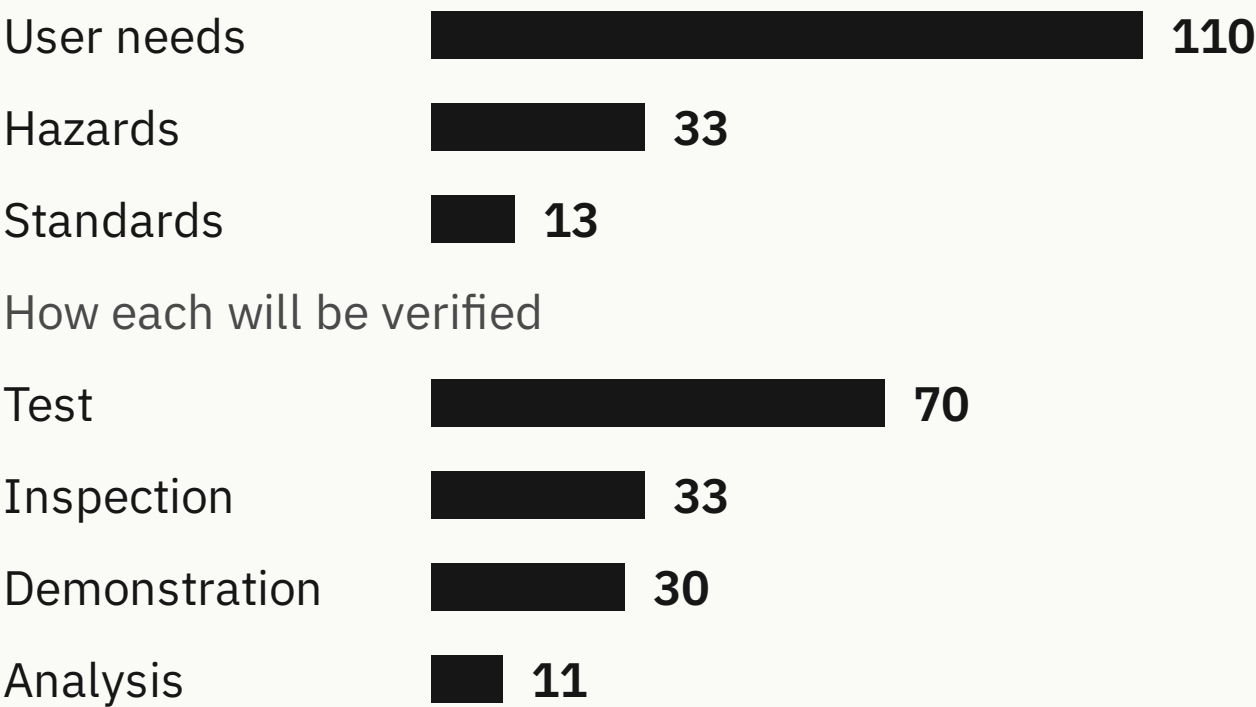
16 user needs refined into 142 requirements

Met

System requirements derived from each user need (SRS-CSI-001 Rev C)



Where the 142 come from



A requirement can have more than one source or method. User needs unchanged since Gate 1.

All 61 findings closed; 3 values still to confirm

Met with action · A2

Findings by type (two reviewers per section, INCOSE writing rules)

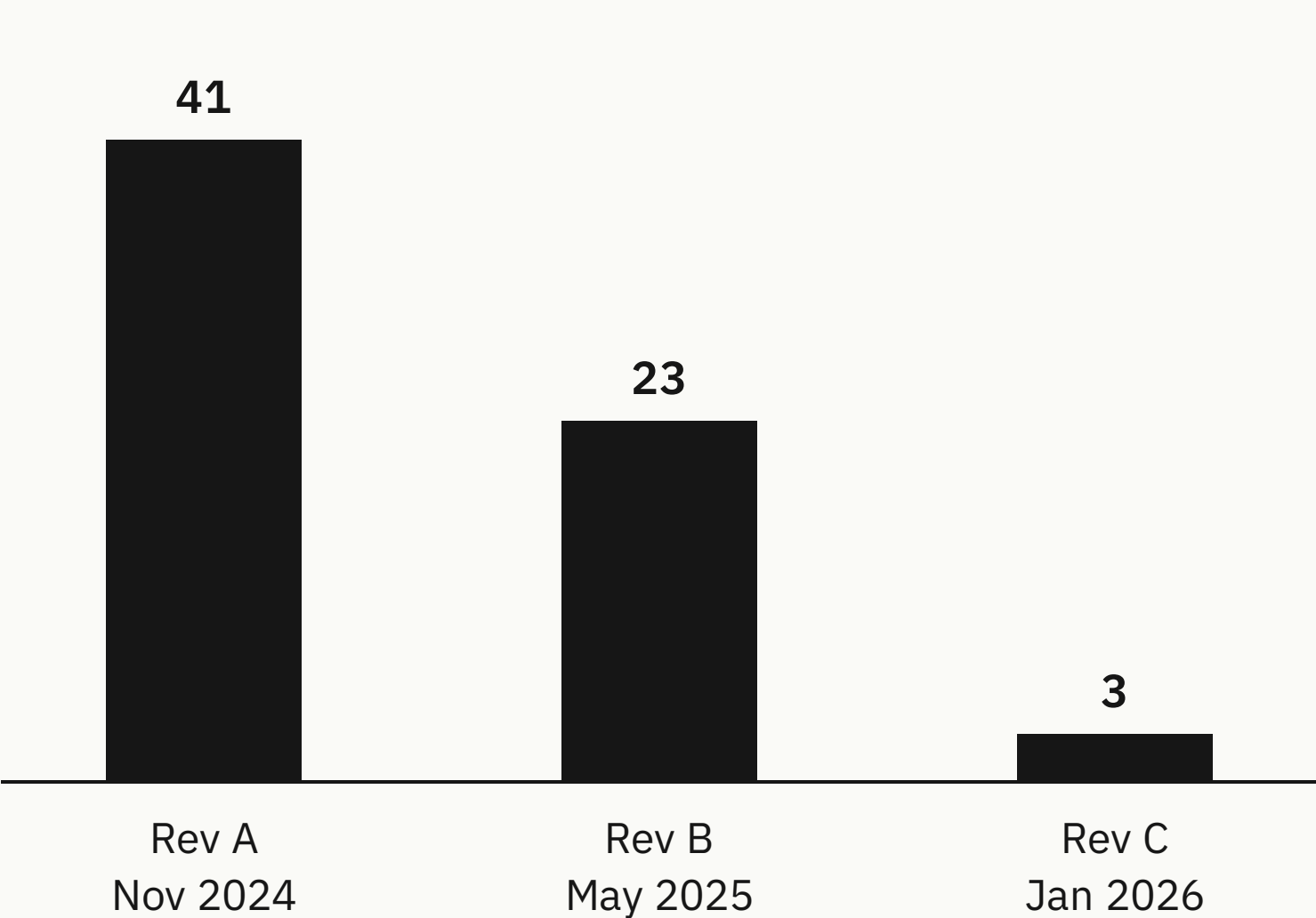


Every finding was corrected in SRS Rev C and re-checked.

Still to be confirmed (TBR) or defined (TBD)

- TBR-01 · SYS-471 carryover limit: 1.0% held, 0.5% target
- TBR-02 · SYS-543 clog response: 5 s held, 2 s target
- TBD-03 · Section 12 transport and installation

Open TBD and TBR items by SRS revision



Eight of nine trace links are complete

Met with action · A1

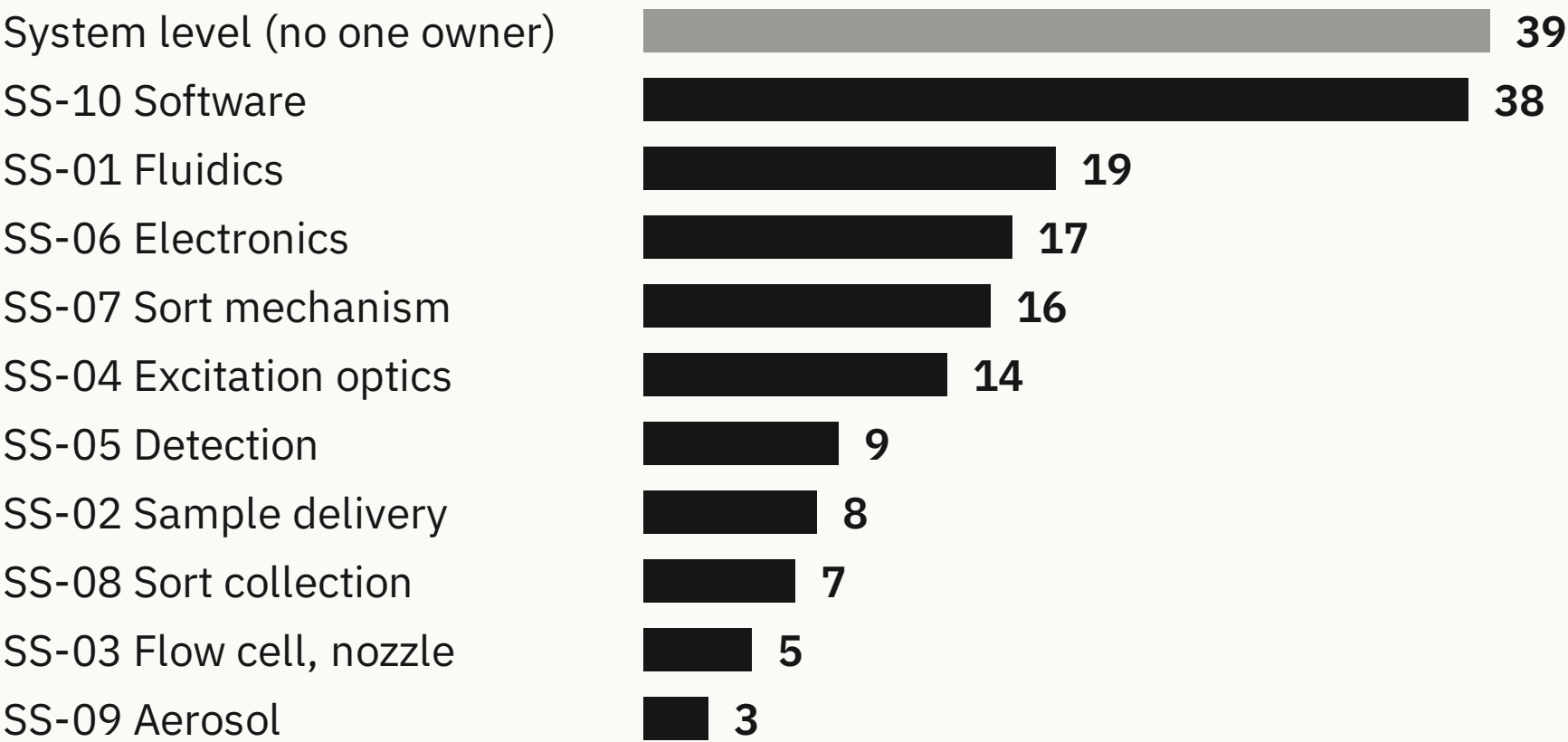
User need → system requirement		16 / 16
System requirement → source		142 / 142
System requirement → allocation		142 / 142
System requirement → verification method		142 / 142
Hazard → control requirement		14 / 14
Software requirement ← SS-10 requirement		29 / 38
Software requirement → parent		131 / 131
Software requirement → software item		131 / 131
User need → planned validation study		16 / 16

The gap is SYS-701 to 709 (Part 11 and cybersecurity), added in SRS Rev C this month. Links to design outputs and test protocols are Gate 3 criteria.

Every requirement has a home in the architecture

Met with action · A3, A5

System requirements allocated to each subsystem



Shared requirements count once per subsystem. Interfaces: 7 of 8 ICDs released; fluidics umbilical ICD-CSI-003 in draft (A5).

Software baseline

131 software requirements, 12 software items, 14 SOUP components.

IEC 62304 Class B

Laser, pressure and motion hazards are stopped by hardware; wrong results by QC and assay controls.

Condition (A3)

Only the controller firmware removes high voltage today. An independent hardware safety supervisor is added in SAD Rev B.

Every hazard is controlled by requirements

Met with action · A4

Hazard	Initial risk	Control requirements
HA-04 Infectious aerosol	Unacceptable	5
HA-01 Laser radiation	Unacceptable	4
HA-02 High voltage	Unacceptable	4
HA-08 Incorrect result	Unacceptable	4
HA-07 Wrong sort output	Unacceptable	2
HA-05 Sort chamber access	Unacceptable	1
HA-15 Fluid run-out	Unacceptable	1
HA-12 Mains, fire	Review	4
HA-09 Stored pressure	Review	2
HA-14 Misconnection	Review	2
HA-13 EMC	Review	1
HA-06 Moving parts	Acceptable	5
HA-10 Chemicals	Acceptable	1
HA-11 Mass, stability	Acceptable	1

14 / 14

hazards with at least one control requirement

33

system requirements are risk controls

Open (A4)

Security threats not yet a hazard source.

Six risks, each with an owner

Met



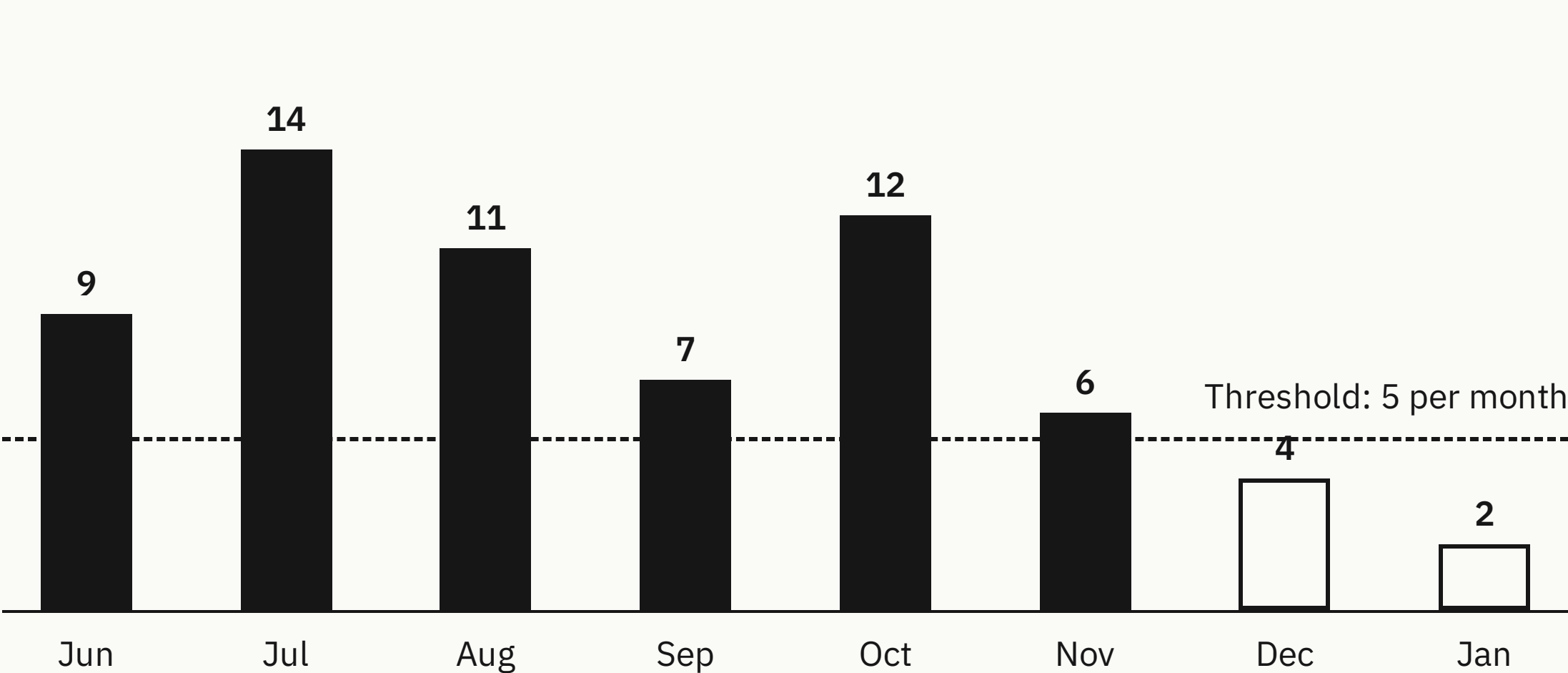
Rows: likelihood · Columns: impact

ID	Risk and mitigation	Owner	Rating
PR-01	Single laser supplier; second source and change notice clause (A6)	Supply Chain	M / H
PR-02	Purity at 70 μm, 4-way sort; margin study Apr 2026	R&D	M / H
PR-03	Options add 6 weeks; separate work package	Program	H / M
PR-04	Late Part 11 and security scope; A1, SAD Rev B	Software	M / M
PR-05	Thermal margin, 6 lasers at 30 °C; confirm on DV	R&D	L / H
PR-06	IVDR notified body capacity for assays	Regulatory	M / M

Changes are below threshold for two months

Met

Approved requirement change requests per month, Jun 2025 to Jan 2026



Rev B (117) to Rev C (142)



The October peak is the option scope decision (aerosol, plates, temperature). Outlined bars are under threshold.

Seven actions, all due before Gate 3

0 blocking

Action · owner	Feb	Mar	Apr	May	Jun	Jul	Aug	G3
A1 Software flow-down of SYS-701 to 709 · P. Nair	27 Feb							
A5 Release umbilical ICD-CSI-003 · J. Whitfield	20 Mar							
A3 Independent HV safety supervisor · P. Nair	31 Mar							
A4 Security threats in hazard analysis · M. Serrano	30 Apr							
A6 Laser second source, change notice · Supply	30 Jun							
A7 Draft V&V plan VVP-CSI-001 · V&V lead	31 Jul							
A2 Close TBR-01, TBR-02, TBD-03 · A. Lodhi	14 Aug							

Blocking items: two found in pre-review (an interlock requirement with no verification method; conflicting HV limits), both fixed in SRS Rev C.

Gate 3 requires: no TBD or TBR, every requirement traced to a design output and a test protocol, and A1 to A7 closed.

Go: exit Phase 2, start Development

No criterion is unmet and nothing blocks detailed design.

Freeze the baseline

SRS Rev C and HA-CSI-001 Rev B go under design change control

Track the conditions

A1 to A7 reported at each monthly program review

Next gate

Phase 3 exit, the verification baseline, August 2026